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Clean-coast index—A new approach for beach cleanliness assessment

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Abstract

A clean-coast index (CCI) was developed and is suggested as a tool for evaluation of the actual coast cleanliness. It measures plastic debris as a beach cleanliness indicator, in an easy way precluding bias by the assessor. Furthermore, the CCI is the measuring tool of the "Clean Coast" program—a new, long-term approach for cleaner beaches by various activities such as an increase in the public awareness. The CCI was proved to be a useful tool for measuring progress and the success of activities such as education campaigns, media coverage and enforcement actions.
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1. Introduction

Marine litter is a growing environmental problem worldwide [1]. It is threatening marine and coastal wildlife [2–8] as well as causing loss of coastline aesthetic quality [9,10]. It has a negative impact on the economy and on health [11–13]. Marine litter is defined as "solid materials of human origin that are discarded at sea or reach the sea through waterways or domestic and industrial outfall" [14].

There are two sources of marine litter: sea-borne and land based—such as industrial wastes and litter originating from human activity on the beach [13,15]. According to the US Department of Commerce and the US Navy, approximately 80% of litter is washed off the land, blown by winds, or intentionally dumped from shore [16]. According to their 2004 cleanup report, over 60% of all debris found during the cleanup originated from recreational activities.

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Israel's Mediterranean coast is characterized by a rich diversity of natural, environmental, cultural and economic resources and is among the country's most valuable assets. The quality of Israel's coast has deteriorated substantially over the last few years as a result of marine litter. Current and wind regimes in the eastern Mediterranean are responsible for deposition of significant quantities of waste from outside the country on the Israeli coast. Still, over half the litter is left behind by vacationers and bathers [28]. Ironically, high litter density, of more than 10 large items/m of beach would deter over 50% of beach users [17].

Litter which reaches the sea, borne by wind or waves, causes damage to the marine biota, and underwater ecosystems. Some of the litter, especially plastic, is returned to some beaches, as a result of storms, as the energy of the waves draws plastic bags out of the seabed.

1.1. "Clean Coast" program

Local authorities in Israel lack cleaning budgets. This, combined with poor habits of the native population who tend to leave their trash behind, has caused litter to pile up mainly on unauthorized beaches, which are out of the sphere of public awareness. Local authorities regularly clean just the authorized bathing beaches, which constitute less than 11% of the Mediterranean coastline (~20 km).

In an attempt to solve the problem of litter on Israel's beaches, in 2005 the Ministry of the Environment launched a long-term program entitled "Clean Coast". The aim of the program is to maintain beach cleanliness at all times, while generating a change in public awareness of the importance of the subject of coast cleanliness.

The program includes several complementary components such as routine cleaning of the coast by local authorities, enforcement against authorities and polluters of the coasts, educational activities in the country's schools and youth movements as well as public relations regarding the program [18].

To estimate the success of the "Clean Coast" Program approach a new and easy tool was needed to assess cleanliness levels of all the country's beaches regularly.

1.2. The clean-coast index (CCI)

For many years there was no accepted index of whether a beach was clean or dirty. The only measure was the amount of waste removed from the beach [16,19]. Success of every cleaning operation was measured by the tonnage of litter cleared from the beach or the number of trash bags collected at the end of the operation, for example UNEP [13], The Ocean Conservancy [16] and Weisskopf [20].

The parameters mentioned above are comparable and easily measured although they do emphasize the cleanup operation, rather than the real cleanliness of the beach. The CCI measures the actual cleanliness of the beach in an objective and easy way precluding bias by the assessor.

2. Methods

As reported earlier in various publications, plastic debris represents by far the majority of litter on the beaches [7,11,21–25].

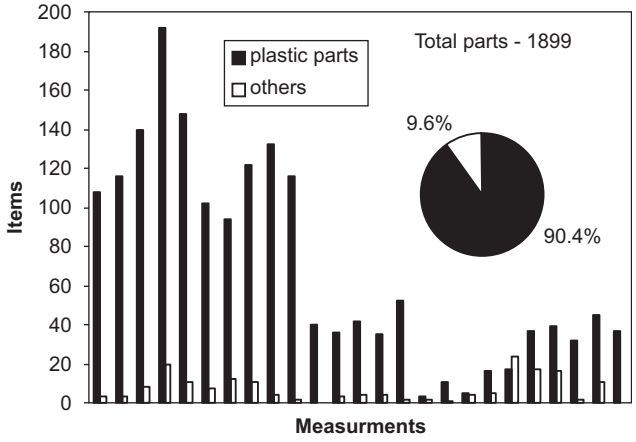


Fig. 1. Plastics and other items counted on 25 beaches at 5 geographical locations.

Preliminary studies conducted throughout the winter of 2004–2005 established these figures as relevant for the Israeli coast as well. According to this study, plastics represent 90% of marine litter, as shown in Fig. 1. In our study, the term 'plastic' refers to any artificial waste made, or partly made, of plastic, including nylon fishing lines, Styrofoam remains, plastic bags in all sizes, polyurethane sheets, bottles and bottle cap, cigarette box outer cover etc. We decided to ignore other waste items, like wood, for instance, that can be counted by some surveyors as waste while others regard it as a natural item. In choosing plastic alone as a numerator, we reasoned that this was similar to the choice of *E. coli* as a marker for human pathogen pollution.

Plastic particles larger 2 cm in size (a bottle cap) were chosen as the index numerator. Furthermore, size was chosen on the basis of practical and measuring convenience as in Ribic et al. [26], and Abu-Hilal and Al-Najar [21].

3. Study area

The Israeli coast stretches about 190 km and it is not homogeneous. It varies from long sandy beaches, surrounded by sand dunes, to narrow rocky shores with beach rocks and pebbled coasts covered by seashells or kurkarite fragments. The eastern Mediterranean is characterized by high-energy waves during summer and winter, and relatively calm sea during spring and autumn. The Israeli coast is not subjected to regular high tides, but shores can be completely submerged by waves when storms occur.

Since litter is affected by the waves, its distribution on a given beach is not constant. After storms, litter is usually found at the highest watermark—sometimes reaching the eastern edge of the beach. When the sea is calm, distribution of litter is affected mainly by vacationers, with higher densities near the waterfront, where people tend to sit. On partly pebbled coasts, the litter, especially smaller items, is caught in stones or shells, whereas sandy areas are generally cleaner. Considering the variety of shore types along the coast, and variations in sea conditions, the cleanliness index should be suitable for all the beaches in Israel, and cover the entire coast.

In agreement with Abu-Hilal and Al-Najar [21], Rees and Pond [15] and Velander and Mocogni [27], the CCI is measured in transects. Transects are performed from the water's edge at the measuring moment, to the border of the coast, represented by any obstacle—sand dune, cliff, vegetation, road or fence.

In order to achieve optimal characterization of beach cleanliness, we chose to perform transects of 10 m in width. The preliminary study established that several 10 m transects, performed on a single beach, gave similar results. Therefore, 10 m transects reliably represent the cleanliness of the beach. To facilitate the counting procedure, transects were divided into 5 strips of 2 m in width. All the unauthorized beaches were morphologically defined as 26 segments, each characterized by the same coastal conditions (sandy/graveled, narrow/wide, open/bordered by cliffs, etc.).

To eliminate bias, the exact measurement location point was not defined and the surveyors were instructed to choose the measurement location randomly before entering the beach (for example by choosing a number of footsteps), prior to knowledge of its cleanliness. Once reaching the location, the exact point was chosen as representative of the beach cleanliness.

The calculation of the CCI is presented in the following equation:

$$\frac{\text{Total plastic parts counted in } Z \text{ lines}}{Z \times 2[\text{m}] \times \text{beach width [m]}} = \text{Plastic parts/m}^2.$$

In order to make the picture clearer for the public, results for appearance of litter on the coasts were graded as follows:

0–0.1 parts/m^2—very clean—no litter is seen
0.1–0.25 parts/m^2—clean—no litter is seen over a large area
0.25–0.5 parts/m^2—moderate—a few pieces of litter can be detected
0.5–1 parts/m^2—dirty—a lot of waste on the shore
More than 1 part/m^2—extremely dirty—most of the shore is covered with plastic debris.

For statistical reasons, as well as for convenience, coefficient $K = 20$ was inserted into the equation. Final CCI numbers are as follows:

0–2: very clean—no litter is seen
2–5: clean—no litter is seen over a large area
5–10: moderate—a few pieces of litter can be detected
10–20: dirty—a lot of debris on the shore
20 +: extremely dirty—most of the beach is covered with plastic debris.

Coast index	Very clean	Clean	Moderate	Dirty	Extremely dirty
Numeric index	0–2	2–5	5–10	10–20	20+

Realizing that the definition of "clean" can differ from person to person we have kept the comparison numerical.

4. Results and discussion

4.1. Baseline index

As mentioned above, the purpose of the index is to measure beach cleanliness and the success of the "Clean Coast" program. In order to evaluate progress of the program, a baseline index was established, before initiation of the program.

In the first two weeks of June 2005, inspectors of the Marine and Coastal Environment Division in the Ministry of the Environment performed 35 CCI measurements on 26 unauthorized beaches along the Israeli Mediterranean coast. On each beach, 1–3 measurements were made, depending on the length of the beach. Baseline CCI results are presented in Fig. 2.

It should be noted that the bathing season opened during the month of April, so that some local authorities had already begun cleaning some of the beaches prior to the beginning of the program.

4.2. CCI 2005

The CCI was measured every 2 weeks. Initially, measurements were conducted at 39 locations on 26 beaches, but due to operational demands, 25 measurements were added. Thus throughout most of the season the index was measured at 59 locations on 39 beaches.

The CCI was measured from June through December 2005 by more than 60 people: inspectors from the Ministry of the Environment and the Nature and Parks Authority, as well as by volunteers of different backgrounds and ages. The assessors reported that each measurement took as long as 20 min and was easily performed. Feedback from all assessors was in agreement with our coast grading: CCI results reflected actual cleanliness of the beaches measured very well.

CCI data collected from June to December 2005 are presented in Table 1. Fig. 3 summarizes the 776 CCI measurements made throughout 2005 at the different beaches.

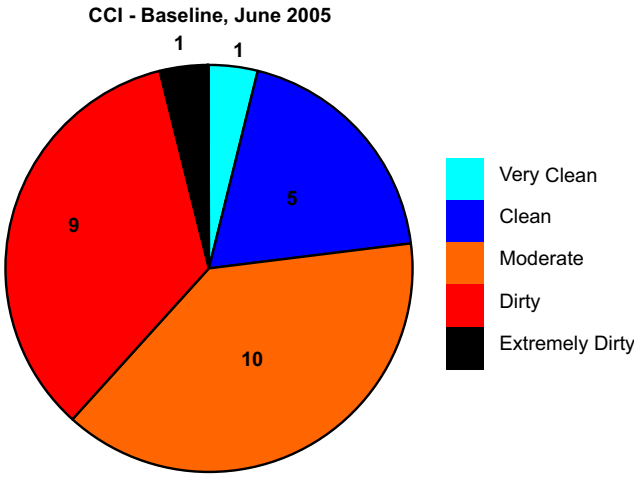


Fig. 2. Results of baseline CCI taken at the beginning of "Clean Coast" program.

Table 1
CCI data collected form June to December 2005

Beach/date	1/6	26/6	21/7	29/7	4/8	18/8	5/9	20/9	3/10	19/10	2/11	15/11	1/12	13/12	27/12
Rosh Hanikra	3.58		5.76	1.25	7.47	6.12	2.17	3.69	6.00	3.11	4.58	8.90	9.88	2.65	3.10
Betzet	6.26		2.55	8.56	1.64	5.83	1.71	2.00	4.83	4.00	5.26	9.83	6.00	3.64	2.26
Kziv	14.48		4.33	4.13	0.75	4.31	2.67	1.79	2.53	1.56	4.37	4.29	6.00	0.93	3.14
Achziv	12.24		2.12	4.07	2.73	1.82	2.61	2.07	3.50	2.26	2.80	1.38	2.30	0.73	1.22
Naharia N.	12.24	4.14	1.33	1.93	1.10	0.93	1.75	1.07	1.00	0.72	0.97	4.07	0.90	0.80	1.81
Naharia C.	5.13	6.17	4.43	2.50	3.40	1.56	0.91	1.74	0.38	0.77	1.30	2.53	2.00	2.60	1.89
Naharia S.	12.65	0.75	1.71	1.58	0.78	1.05	3.25	0.81	2.20	2.09	1.67	1.83	1.42	1.20	1.89
Shavey Zion	5.27			2.00	1.58	1.85	4.67	2.30	2.63	1.79	3.60	4.36	6.23	4.00	5.12
Hatzrot Yasaf	14.05	12.01	2.50	2.08	2.08	2.95	7.78	5.50	6.53	7.77	2.85	6.22	10.15	6.67	5.47
Bustan Hagalil	11.49	33.38	7.86	1.57	2.42	3.62	8.67	5.83	11.93	11.40	4.30	22.07	11.60	10.30	5.68
Acre N.	11.99	6.61	36.89	20.00	20.55	8.00	3.39	1.29	0.60	4.21	4.59	6.22	3.73	3.84	3.55
Acre S.	6.52		8.04	16.56	18.43	18.14	1.14	8.18	1.68	5.83	2.87	3.95	2.87	1.90	1.82
Kiryat Yam N.	11.62	16.30	17.68	17.00	10.61	9.58	12.64	10.44	0.75	1.56	9.39	5.93	3.53	4.43	4.39
Kiryat Yam S.	11.62	16.30	17.68	17.00	10.61		4.23		1.69	1.93	2.29	1.26	2.00	1.89	1.49
Kiryat Hiam	10.68		13.78		10.16	13.73	16.85	18.25	3.36	9.66	5.83	16.16	8.28	7.96	23.27
Shikmona			4.53			7.65	11.83	17.05	4.40	14.29	7.43	7.58	7.69	7.91	6.00
Dado N.	11.31		4.12				4.65	17.54	2.76	1.12	0.89	2.15	3.74	6.04	8.06
Tirat Hacarmel	4.41		3.90		3.50	2.56	4.11	6.80	3.42	2.35	0.75	2.39	1.50	2.37	3.28
Hof Carmel N.	2.43		1.74			7.14	11.91	5.07	2.74		8.78		6.71	1.95	4.48
Hof Carmel S.	2.43								8.06	8.83				5.60	3.08
Neve Yam N.	1.51		3.25	1.41		2.44	2.34	0.55		1.52	2.19	2.00	2.64		0.93
Neve Yam S.	8.10		4.56			3.97	1.51	0.36			0.88	4.34	0.75		1.50
Habonim	1.14				1.61	3.28	1.56	1.49	3.14	1.94	3.73	2.86	0.58		1.40
Nachsholim	9.46		1.69	2.37	2.21		15.23		21.82	11.00	8.11	10.29	1.03		
Dor	2.31		3.50	2.50	2.90	1.22	1.46	1.67	1.73	1.85	1.29	2.06	1.88	1.74	1.15
Maayan Zvi	1.74		2.00	2.57	2.24	0.57	1.05	0.80	1.10	2.17	1.60	1.89	0.85	1.78	1.68
Maagan Michael	4.56		2.74	3.47	1.48	0.79	1.29	2.30	0.93	1.67	1.40	1.94	1.25	1.25	1.23
Jiser A-Zarka	7.92	9.11	4.40	4.48	3.93	7.08	3.64	3.11	3.29	2.34	1.36	4.00	1.79	3.79	3.66
Cisaria	8.98	0.89	2.18		2.08	0.83				5.70	5.31		10.27	2.46	
Sdot Yam	5.21	4.81	3.87	7.43	4.92	3.88	7.28	6.58	3.86	4.31	4.36		3.10	3.18	
Hadera	14.31	9.50	4.17	5.24	8.83	5.63	3.48	3.48	1.13	3.10	2.74	0.09	0.70	0.87	5.66

Table 1 (*continued*)

Beach/date	1/6	26/6	21/7	29/7	4/8	18/8	5/9	20/9	3/10	19/10	2/11	15/11	1/12	13/12	27/12
Gdor	7.26	29.80	2.62	9.56	7.78	13.06	5.75	25.57		3.83	4.67	3.27	5.21		2.90
Michmoret	3.22		3.60	11.44	6.69	14.56	18.95	2.39		3.00	0.92	21.69	13.78		13.36
Beit Yanay	1.02	3.24		2.86	2.59	4.95	2.88	4.11	1.88	2.94	7.44	2.72	1.07	3.20	6.25
Neurim	6.04	7.00	1.71	3.57	9.40	8.95	6.22	11.10	4.84	9.65	9.33	2.38	2.90	1.14	2.90
Emek Hefer	12.11	10.00	7.86	14.77		6.59	10.25			1.04	13.78	24.82		2.97	24.25
Natania N.	4.41	7.09	5.00	13.20	6.14	3.80	2.55	4.13	3.40	1.70	9.73	12.24	4.00	3.40	4.09
Natania S.	4.41		8.33	5.45	8.00	15.18	7.07	5.75	14.67	3.89	6.38	8.42	3.94	4.95	15.38
Poleg	12.53	9.90	3.27	11.26	5.68	5.23	6.00	2.64	4.03	4.33	2.73	1.29	3.48	4.92	
Hof Hasharon	19.16	57.25	12.71	21.14	13.06	10.00	4.29	14.15	2.75	5.88		44.50	21.00	19.33	19.78
Gaagh	36.29	17.78	13.26	22.11	13.76	5.06	9.88	3.83	4.00	3.88		36.27	22.00	4.13	1.68
Apolonia	6.80		32.34		24.41		14.06	10.67	20.40	10.94		5.18	3.20		
Herzelia S.					5.44		3.07	0.80	1.56	1.50	2.67	4.63	2.29	1.13	
Hatzuk	5.60		9.40		9.60	4.59	5.00	2.60	2.08	2.56	6.36	9.23	24.00	7.76	
Tel Baruch	17.95		7.69		10.58	7.89	5.81	3.32	4.00	4.40	3.39	2.50	17.14	3.22	
Bat Yam			0.45			1.48	3.07	1.59	4.20		3.90	8.73		2.70	5.75
Palmachim N.	3.00		4.44			7.60	21.10	16.13	8.92	3.20		5.40	19.36	3.96	1.20
Gan Rave			2.33	6.43		6.92	6.11	3.85	5.39	6.71	2.91	3.71	1.42	4.16	2.02
Palmachim S.	28.93		8.28	9.41		8.82	1.07	3.28	26.13	3.08	3.99	0.80	9.38	4.39	18.81
Ashdod N.	6.53		7.38				3.58	2.04		14.93	4.05	3.96	1.79	7.91	2.21
Ashdod S.	7.42		9.31	4.71			22.22	2.50	12.27	11.95	4.53	1.93	2.40	5.94	9.85
Beer tuvia	9.84			6.07		6.15	9.88	3.24		2.21	4.70	6.70	5.49	6.98	9.85
Nitzanim	6.70			4.22			6.57	5.52		9.13	3.25	8.20	3.00	2.88	9.40
Avtach	6.21		5.82	18.91		6.94	4.80	20.00	19.12		5.93		9.11	3.19	4.54
Ashkalon N.	4.46		1.10	13.78			10.13	18.00		14.35	18.74	7.18	3.33	2.53	3.36
Gan Leumi Ashkalon	10.21		8.29			5.54	7.38		17.76	13.80	21.11	7.62	6.91	4.00	4.16
Ashkalon S.	20.44		15.18	8.70			7.43	19.73	18.80	16.21	4.57	6.77	9.74	1.65	1.11
Zikim N.	15.60		18.08	11.65		11.47	5.52	9.53	10.42	8.11	20.43	13.37	4.57		1.33
Zikim S.	10.32		17.19	29.29		37.39	3.72							2.43	

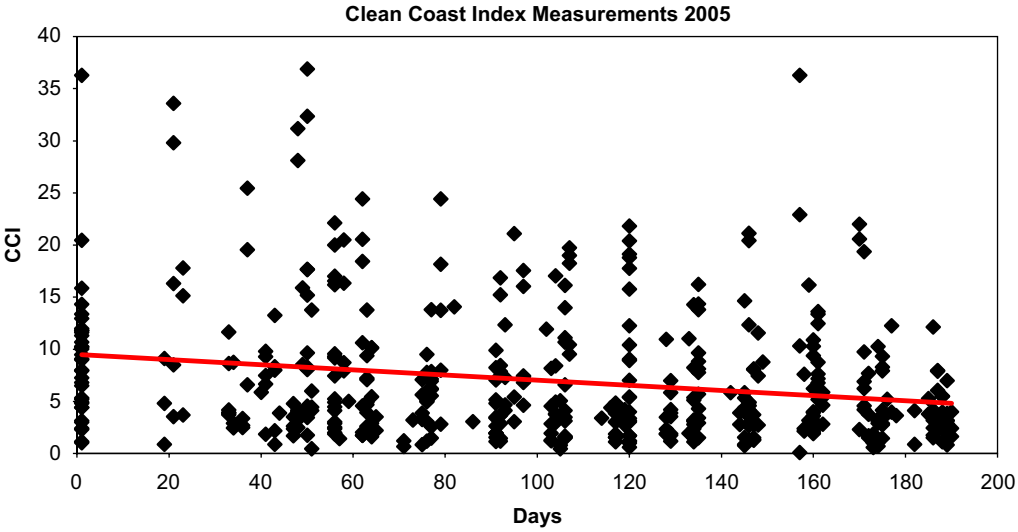


Fig. 3. Total CCI measurements performed from June through December 2005.

The data demonstrate a mild trend line, declining as the season progressed, as the cleanliness of the coast increased.

4.3. CCI implementation

The CCI was used to increase the public awareness regarding coast cleanliness and motivate the authorities to clean their beaches. It has been published regularly both in the Ministry of the Environment's web site 〈www.environment.gov.il〉 and the Israel's national and local newspapers since June 2005.

The CCI publications had great impact on the authorities for various reasons.

In local communities along the coast, the beaches conditions are among the most important environmental issues and therefore public awareness and media coverage is wide. Routinely, the local newspapers criticize the authorities about their way of spending the public taxes. The CCI was an important parameter used by the authorities showing cleanliness activities. Thus, coast cleanliness on those authorities is among their main issues and they are willing to join the Clean Coast Program and get financial support to clean their beaches.

In addition to evaluating the progress of the "Clean Coast" program, the CCI has generated another important benefit. It enables better control and regulation of the "Clean Coast" budget that funds most of the cleaning activities conducted by local authorities. The CCI supported initiation of enforcement activities, in case of failure to maintain coast cleanliness in accordance with contracts between local authorities and the "Clean Coast" program.

For example, North Acre beach was not cleaned by the municipality at the beginning of the program. Due to enforcements activities, the municipality joined the project in August.

The change in the CCI was immediate, as can be seen in Fig. 4. The graph compares Acre with Habonim beach, which is part of a national park and is therefore regularly

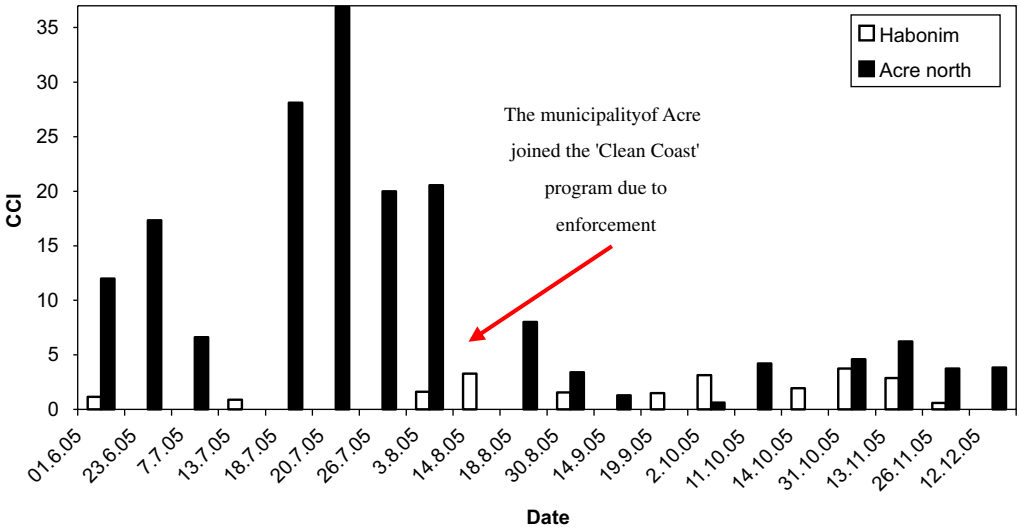


Fig. 4. CCI of Habonim and Acre beaches, June–December 2005.

cleaned by the Nature and Parks Authority (its CCI rating has generally been very clean throughout the year).

After the improvement in the beach cleanliness due to the clean up by local authorities, the CCI publication gave us a platform to motivate the general public for keeping the coast clean, by taking responsibility of their waste.

5. Conclusions

For the first time, an easy and objective means of assessing the cleanliness level of the coast was developed.

The CCI is an accurate measurement instrument for coastal cleanliness. Visual assessments appeared to have a high correlation with the index outcome.

The CCI proved to be a useful tool for measuring the progress and success of different actions that are aimed to improve the beach cleanliness, such as the Clean Coast program. The index and the consistency of measurements constitute a useful tool for strategic and operational resource allocation. It indicates which topics require greater attention: education, media coverage and public awareness, ramping up enforcement and cleaning efforts.

CCI was found to be less effective in extreme cases, such as beaches where the debris was mainly of organic matter (such as can be found near estuaries) or where particles were of extremely large size. Obviously, the index would also prove ineffective if the beach was covered with algae (which lower the aesthetic quality of the beach), tar or charcoal etc. The CCI is less effective when an easterly wind is blowing, because this buries the litter in the sand. Since such occurrences are rare and require immediate intervention we regard this as a minor flaw.

The CCI is a handy tool for evaluation of the success of the "Clean Coast" program, and can be very useful for implementation, with minor adaptation, in other countries, which choose to focus on actual and consistent results.

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References

[1] Ross JB, Parker R, Strickland M. A survey of shoreline litter in Halifax Harbour 1989. Marine Pollution Bulletin 1991;22:245–8.
[2] Auman HJ, Ludwig JP, Giesy JP, Colborn T. Plastic ingestion by Laysan Albatross chicks on Sand Island, Midway Atoll in 1994 and 1995. In: Robertson G, Gales R, editors. Albatross biology and conservation. Surrey Beatty: Chipping Norton; 1998. p. 239–44.
[3] Bugoni L, Krause L, Petry MV. Marine debris and human impacts on sea turtles in southern Brazil. Marine Pollution Bulletin 2001;42(12):1330–4.
[4] Chiappone M, White A, Swanson DW, miller SL. Occurrence and biological impacts of fishing gear and other marine debris in the Florida Keys. Marine Pollution Bulletin 2002;44(7):597–604.
[5] Haynes D. Marine park debris in the far northern section of the Great Barrier Reef Marine Park. Reef Research 1997;7:3–4.
[6] Ryan PG, Connell AD, Gardner BD. Plastic ingestion and PCBs in seabirds: is there a relationship? Marine Pollution Bulletin 1988;19(4):174–6.
[7] Ryan PG, Moloney CL. Plastic and other artefacts on South African beaches: temporal trends in abundance and composition. South African Journal of Science 1990;86:7–10.
[8] Tomas J, Guitart R, Mateo R, Raga JA. Marine debris ingestion in loggerhead sea turtles, *Caretta caretta*, from the western Mediterranean. Marine Pollution Bulletin 2002;44(3):211–6.
[9] EPA. plastic pellets in the aquatic environment: sources and recommendations. EPA 842/B-92/010. USEPA, Office of Water, Washington, DC, USA, 1992.
[10] Somerville SE, Miller KL, Mair JM. Assessment of the aesthetic quality of a selection of beaches in the Firth of Forth, Scotland. Marine Pollution Bulletin 2003;46(9):1184–90.
[11] Alkalay R. All about coastal debris or why the beaches are so dirty. Marine and coast articles, Israeli Ministry of the Environment, 2000:282–5 in Hebrew.
[12] Thompson RC, Olsen Y, Mitchell RP, Davis A, Rowland SJ, John WGA, et al. Lost at sea: where is all the plastic? Science 2004;304:838.
[13] UNEP. Marine litter. An analytical overview. 47, 2005.
[14] National Academy of Sciences, Marine litter. In: Assessing potential ocean pollutants. A report of the study panel on commission on natural resources, Natural Research Council, National Academy of Sciences, Washington, DC, USA,1975.
[15] Rees G, Pond K. Marine litter monitoring programmes-a review of methods with special reference to national surveys. Marine Pollution Bulletin 1995;30(2):103–8.
[16] The Ocean Conservancy. Marine debris: where does it come from? The international Cleanup 2005;2004:2–5.
[17] Ballance A, Ryan PG, Turpie JK. How much is a clean beach worth? The impact of litter on beach users in the Cape Peninsula, South Africa. South African Journal of Science 2000;96(5):210–3.
[18] Alkalay R, Pasternak G, Zask A, Amir R. Clean Coast Program, 2005. report. In preparation.
[19] Dixon TJ, Dixon TR. Marine litter distribution and composition in the North Sea. Marine Pollution Bulletin 1983;14:145–8.
[20] Weisskopf M. Plastic reaps a grim harvest in the oceans of the world. Smithsonian 1988:65–70.
[21] Abu-Hilal AH, Al-Najjar T. Litter pollution on the Jordanian shores of the Gulf of Aqaba (Red Sea). Marine Environmental Resarch 2004;58:39–63.
[22] Caulton E, Mocogni M. Preliminary studies of man-made litter in the Firth of Forth, Scotland. Marine Pollution Bulletin 1987;18:446–50.
[23] Dixon TJ, Dixon TR. Marine litter surveillance. Marine Pollution Bulletin 1981;12(9):289–95.
[24] Ryan PG, Swanepoel D. Cleaning beaches: sweeping litter under the carpet. South African Journal of Science 1996;92:275–6.

[25] UNEP Guidelines. Management of Coastal litter for the Mediterranean region. UNEP (DEC) MED WG. 231/7, 2003.
[26] Ribic CA, Dixon TR, Vining I. Marine Debris Survey Manual, NOAA Technical Report NMFS 108, US Department of Commerce. Washington, DC, USA, 1992.
[27] Velander K, Mocogni M. Beach litter sampling strategies: is there a 'best' method? Marine Pollution Bulletin 1999;38(12):1134–40.
[28] Gabbay S. Onwards toward cleaner coasts. Israel Environmental Bulletin 2005;29:6–8.